Second Voice

A voice in your ear when a phone call turns into a scam.

William Goodwin · Hiya Voice AI Challenge · Real-Time Assistance track

What I built

A working prototype that listens to a practice call and coaches you in your other ear. It names the pressure tactic, gives you a line to say and, once the call has raised a flag, interrupts if you start reading out a code. In scripted practice calls it also backs you up when the caller pushes. It runs in Chrome on your device, with headphones and no API keys. Practice calls and a free-listening mode both use the real speech pipeline.

Who it's for, and the problem

It's for anyone being rushed by a caller who sounds official. In 2025, 17% of UK push payment fraud cases began via telecoms, but made up 28% of losses. That fraud cost £576.4m overall (UK Finance). Hiya and Google can warn you that a call looks like a scam. As far as I could find, they don't tell you what to say next.

Why voice

The phone is at your ear, so you can't see a screen. The caller sets the pace, so there's no time to read. And it can hear you: you can ask "Coach, is this real?" hands-free, and it can stop you reading out a code. After the call a screen is better, so the summary is on screen.

Decisions I'd defend

- **Patterns and a small model classify. Nothing generates speech during a call.** All 35 lines are written in advance, because a caller's speech is untrusted input. A scammer can't talk it into reassuring anyone, and recorded lines need no generating.
- **It listens to behaviour, not voice quality.** A good clone doesn't change what's being asked for, so this complements deepfake detection.
- **Loose similarity matches can warn, but can't order.** A false "hang up" is the costliest mistake, so it takes an explicit ask, a very close match, or signs adding up with at least one matched outright.

What I measured

	Scam: spoke	Scam: hang up	Genuine: spoke	Genuine: hang up
Generated calls no rule was written from (same generator as the tuning data)	159/160	155/160	6/160	3/160
Real calls, the half kept unread	**5/9**	**4/9**	**none in set**	**none in set**
Tuning calls, synthetic voices, simulated phone line	86/90	74/90	2/90	0/90

Real calls are the weak spot. Rules written after reading half the YouTube calls took that half from 5/11 to 10/11. The half I kept unread ended where it began (5/9 spoke, 4/9 hang up), and I've scored it three times. Those rules overfitted, and the figure to trust is about half, from a small sample. False warnings on generated genuine calls did fall, from 15/160 to 6/160. In scripted calls, urgent lines start about half a second after the chunk of speech that caused them. In unbroken speech, one took 2.3 seconds.

AI reviewers found bugs my tests had missed, including a freeze on pressing Hang up during a warning. The freeze is fixed and tested, and what's left is in the README. I shortened the coach's lines and tested them blind, but the reviewers were AI models. **Not tested:** real voices, real phone audio or real users. A browser can't reach call audio. As far as I know, nor can an ordinary iOS or Android app on a call it isn't carrying. A real version needs the call routed to it by a carrier, a phone maker or call forwarding, which Hiya AI Phone appears to use.

What I'd do next

1. Test with real voices, then with the people most targeted.
2. Get real call data. Rules written from 11 calls didn't carry over, so more rules from so few calls won't fix this.
3. Try a language model only as a classifier, limited to the fixed tactics. It still wouldn't write what the coach says.
4. Work out how the call reaches the coach, and how only you hear it. I'd ask Hiya how AI Phone gets its audio.